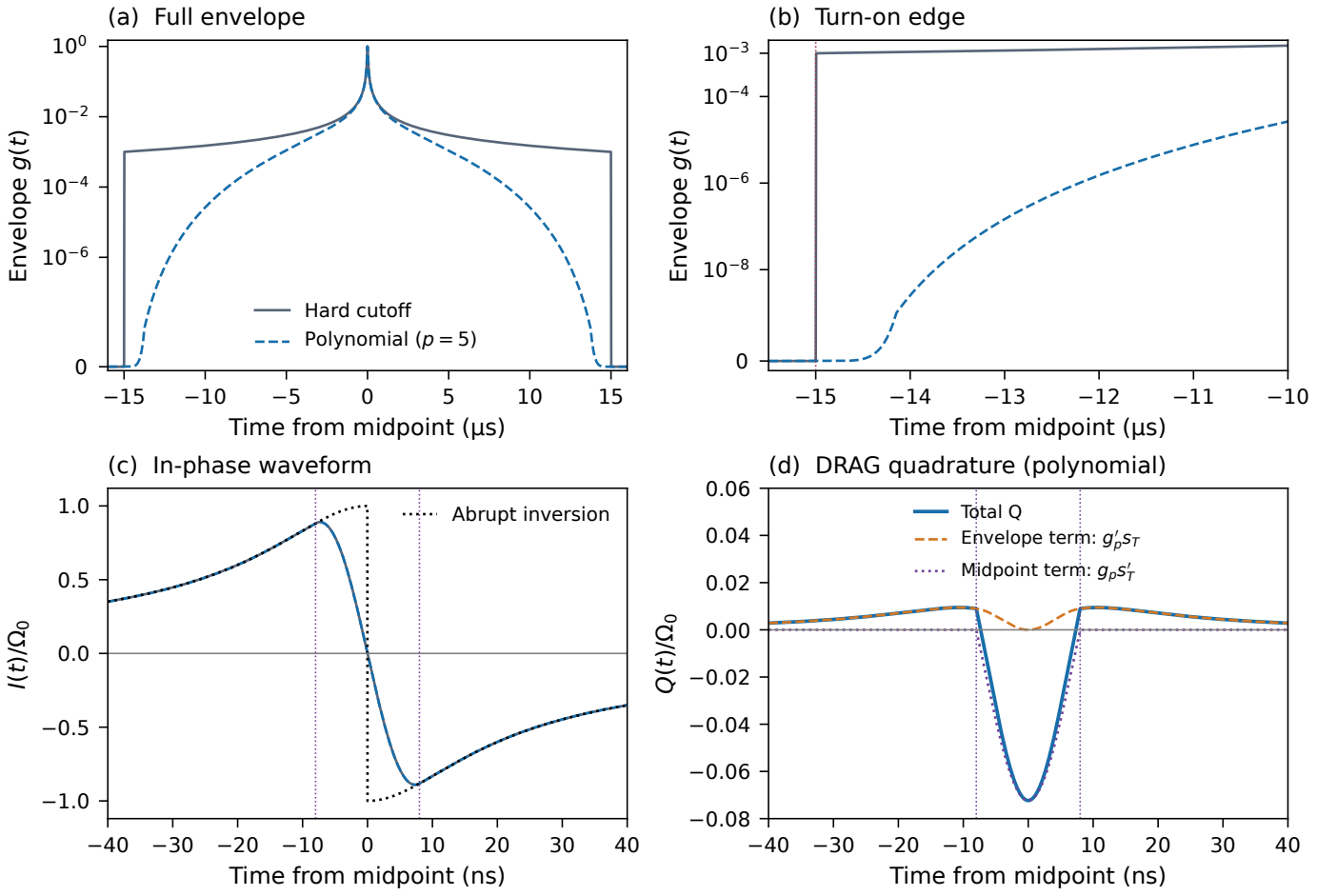
(a) Full envelope
Envelope $g(t)$
10^{0}
10^{-2}
10^{-4}
10^{-6}
0
Hard cutoff
Polynomial ($p = 5$)
−15 −10 −5 0 5 10 15
Time from midpoint (μs)
(b) Turn-on edge
Envelope $g(t)$
10^{-3}
10^{-4}
10^{-6}
10^{-8}
0
−15 −14 −13 −12 −11 −10
Time from midpoint (μs)
(c) In-phase waveform
$I(t)/\Omega_0$
1.0 0.5 0.0 −0.5 −1.0
Abrupt inversion
−40 −30 −20 −10 0 10 20 30 40
Time from midpoint (ns)
(d) DRAG quadrature (polynomial)
$Q(t)/\Omega_0$
0.06 0.04 0.02 0.00 −0.02 −0.04 −0.06 −0.08
Total Q
Envelope term: $g'_p s_T$
Midpoint term: $g_p s'_T$
−40 −30 −20 −10 0 10 20 30 40
Time from midpoint (ns)